

Annexure-I

1. INTRODUCTION:

Rivers play a vital role in sustaining ecosystems, supporting agriculture, and providing water resources for millions of people worldwide. Understanding the dynamics of river discharge—the volume of water flowing through a river channel over time—is essential for effective water resource management, flood prediction, and environmental protection. Traditionally, river discharge is measured at gauging stations using in-situ sensors and manual observations. However, such stations are often sparse, especially in remote or difficult-to-access regions, leading to significant gaps in hydrological data.

To address these challenges, the integration of satellite remote sensing and advanced hydrodynamic modeling has emerged as a promising solution. This approach enables the estimation of river discharge at "virtual stations," which are locations where direct measurements are unavailable. By leveraging high-resolution satellite data and sophisticated modeling techniques, researchers can generate continuous, spatially extensive discharge estimates that inform decision-making and scientific research.

We came up with a comprehensive methodology for estimating river discharge at virtual stations using hydrodynamic modeling and satellite observations, with a case study focused on the Ganga river stretch between Prayagraj and Varanasi. The workflow employs ICESAT-2 satellite data and the Non-Linear Slope Break (NLSB) model, supported by automated algorithms and statistical techniques to enhance accuracy and scalability.

2. STUDY AREA:

The methodology will be tested in selected virtual gauges of Mahanadi and Ganga river basin, once validated, same framework can be applied to other river basins covering more virtual gauges.

Initially, the selected study area is the Ganga river segment between Prayagraj and Varanasi. This stretch is of particular interest due to its hydrological complexity, cultural significance, and the presence of major urban centers. The Ganga river, one of the largest and most important rivers in India, exhibits dynamic flow patterns influenced by monsoonal rainfall, upstream water management, and varying channel morphology.

The primary data source for this study is the ICESAT-2 satellite mission, which provides precise elevation measurements using laser altimetry. The satellite's Advanced Topographic Laser Altimeter System (ATLAS) captures detailed water surface elevations and river widths, which are critical for estimating river cross-sections and, subsequently, discharge. The use of satellite data overcomes the limitations of ground-based measurements, offering frequent, consistent, and spatially comprehensive observations.

3. METHODOLOGY:

In the proposed proposal, the entire methodology proposed in this project is divided into two parts. The first part focuses on all the steps involved in setting up the HD model to simulate water level and rating curves at virtual stations falling within the modelled river reach. The second part deals with the water level retrieval using satellite altimetry data for multi-site validation of the HD model and validation of the rating curves. Later, all the ECVs will be derived at virtual stations using multiple simulations.

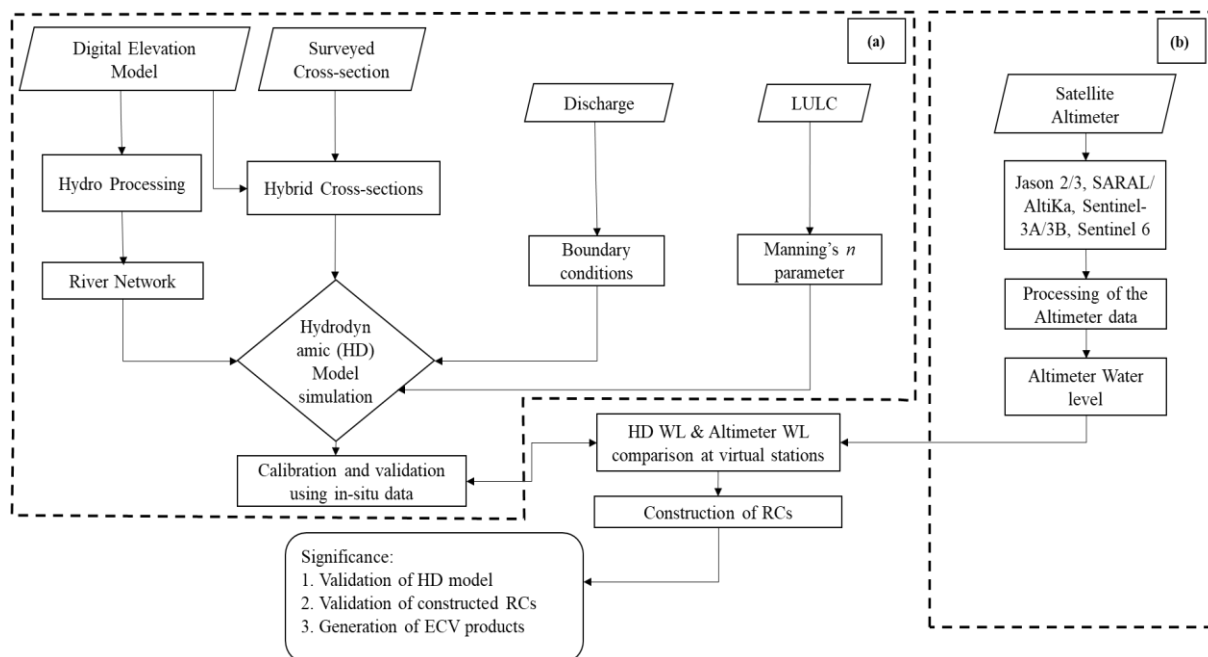


Fig. 1. Methodology

3.1 DATA USED:

Type of data	Data	Source	Temporal /Spatial Resolution
Satellite Altimetry	1. Jason 2 2. Jason 3 3. Saral/AltiKa 4. Sentinel-3A/3B 5. ICESAT ½ 6. SWOT	1. AVISO 2. AVISO 3. AVISO/SAC-MOSADC 4. Copernicus 5. Open altimetry 6. AVISO	1. 9.91 days/ 20 km footprint 2. 10 days / 20 km footprint 3. 35 days / 8 km footprint 4. 27 days / (300 x 1.64) km footprint 5. 90 days / 13-60 m 6. 21 days / 10 km
Digital elevation model	Cartosat / Copernicus/ ALOS PALSAR RTC	NRSC, ISRO / Copernicus / Alaska Satellite Facility	30 m / 30 m / 12.5 m

Optical *	Sentinel 2A/2B Landsat 8 Resourcesat-2/2A	Earth Explorer / Copernicus Earth Explorer / Copernicus NRSC, ISRO	6 days / 10 m 16 days / 30 m 5-24 days / 5.8-56 m
SAR*	Sentinel 1A/1B	Earth Explorer / Copernicus	6 days / 10 m
Land Use and Land Cover (LULC)	-	ISRO GBP NRSC	1:2,50,000 1: 50, 000

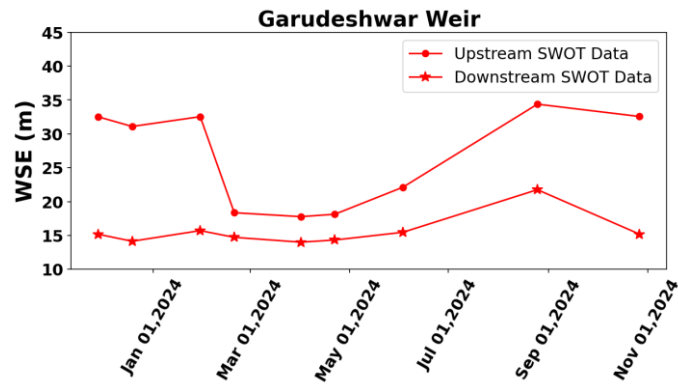
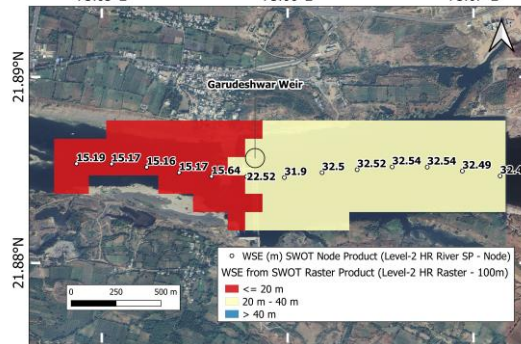
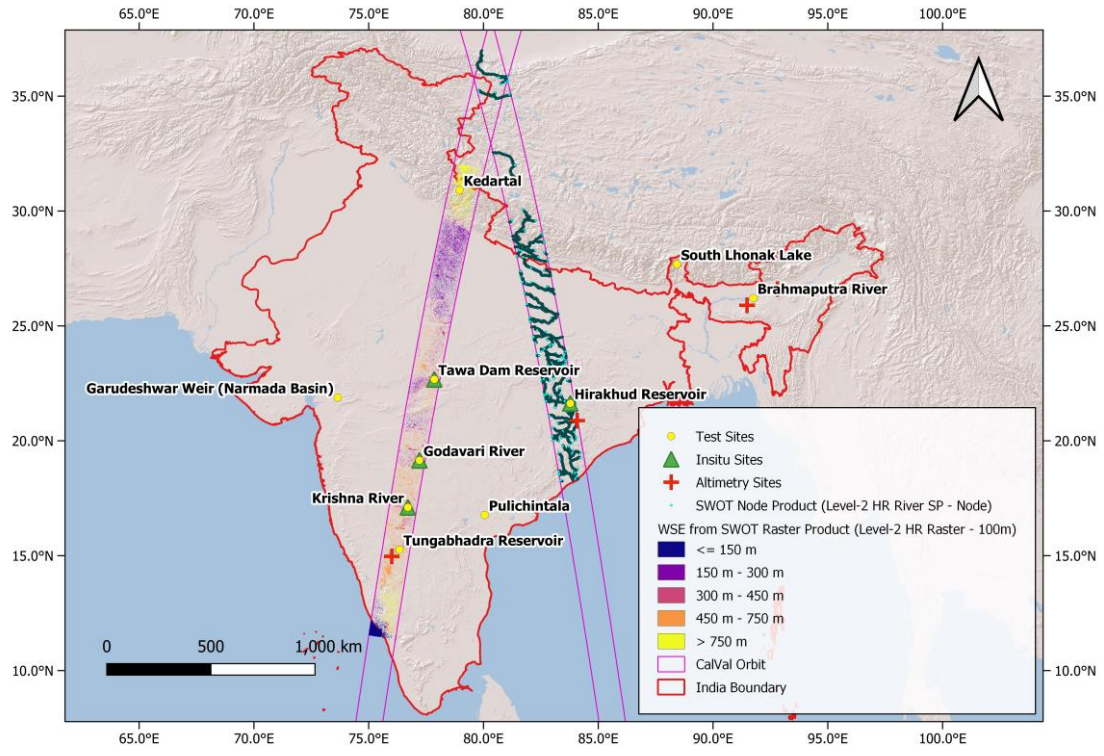
3.2 PROGRESS IN LAST ONE YEAR AND MAJOR ACHIVEMENTS:

3.2.1. Water Level Time Series Extraction:

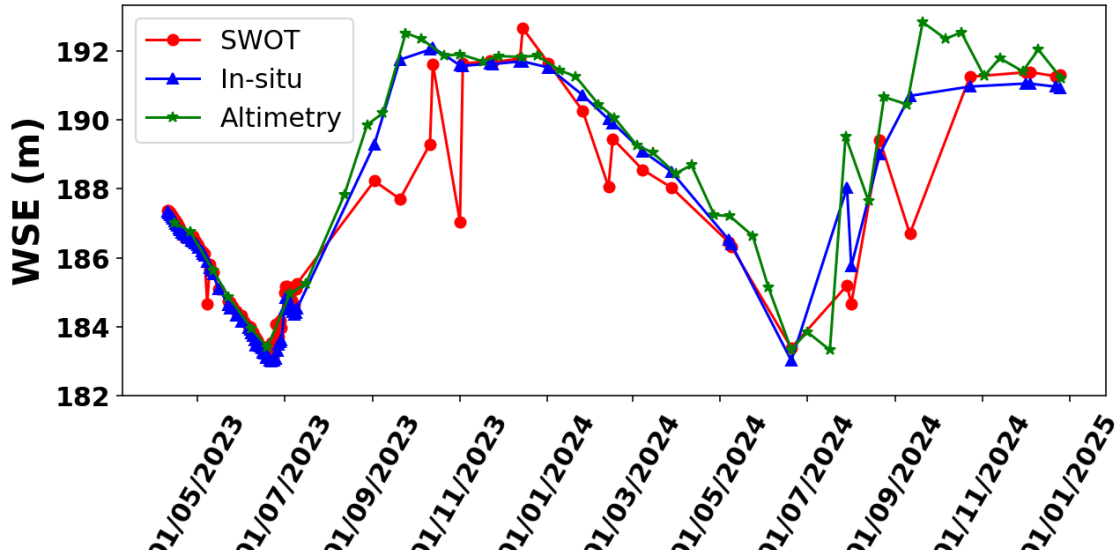
Satellite altimetry has been widely used for assessing water level calibrating hydrological-hydrodynamic models and characterizing floodplains. Various approaches have been developed to estimate river discharge from remotely sensed observations (water level, width, slope, reflectance) most of which are dependent on ground-data and calibration of the models.

Water Surface Elevation (WSE) time series were considered for with insitu-data, historical altimetry missions and the recent Surface Water and Ocean Topography (SWOT) mission by constructed at test sites by digitizing polygons around gauge/virtual stations and lake areas using QGIS Google Map services. For SWOT-derived WSE, we only considered nodes classified as “good” (0) or “suspect” (1) in the quality filter (node_q) were used, while lower-quality data was excluded. Outliers exceeding twice the standard deviation of the time series were removed, though geolocation errors or misidentified sandbanks in nadir altimetry were manually filtered. All water levels were referenced to the EGM 2008 datum.

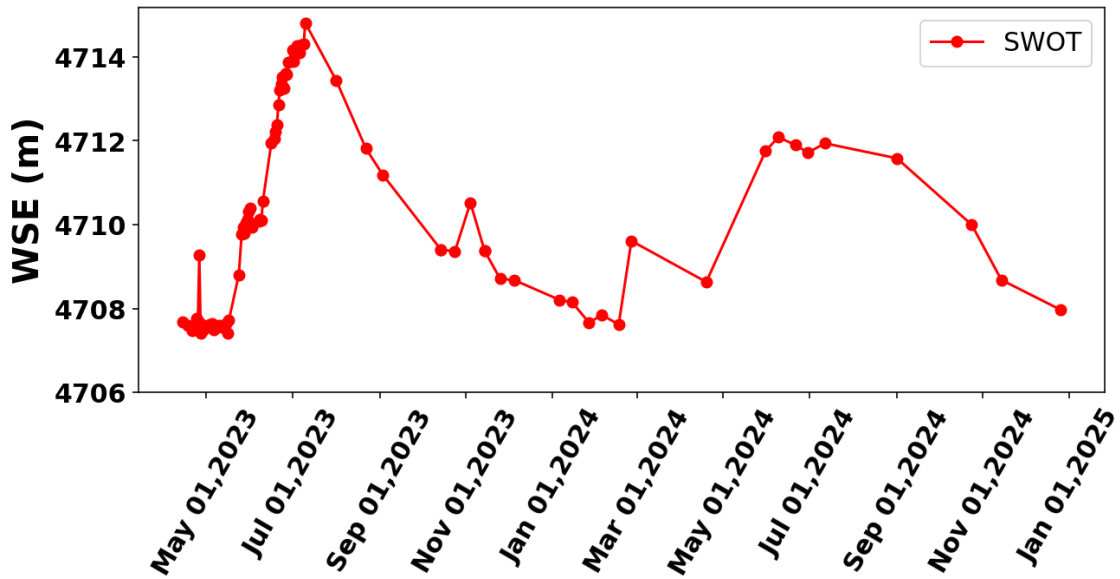
भारत सरकार GOVERNMENT OF INDIA
अन्तरिक्ष विभाग DEPARTMENT OF SPACE
भारतीय अन्तरिक्ष अनुसंधान संगठन INDIAN SPACE RESEARCH ORGANISATION
भारतीय सुदूर संवेदन संस्थान, देहरादून INDIAN INSTITUTE OF REMOTE SENSING, DEHRADUN
सामान्य प्रशासन GENERAL ADMINISTRATION

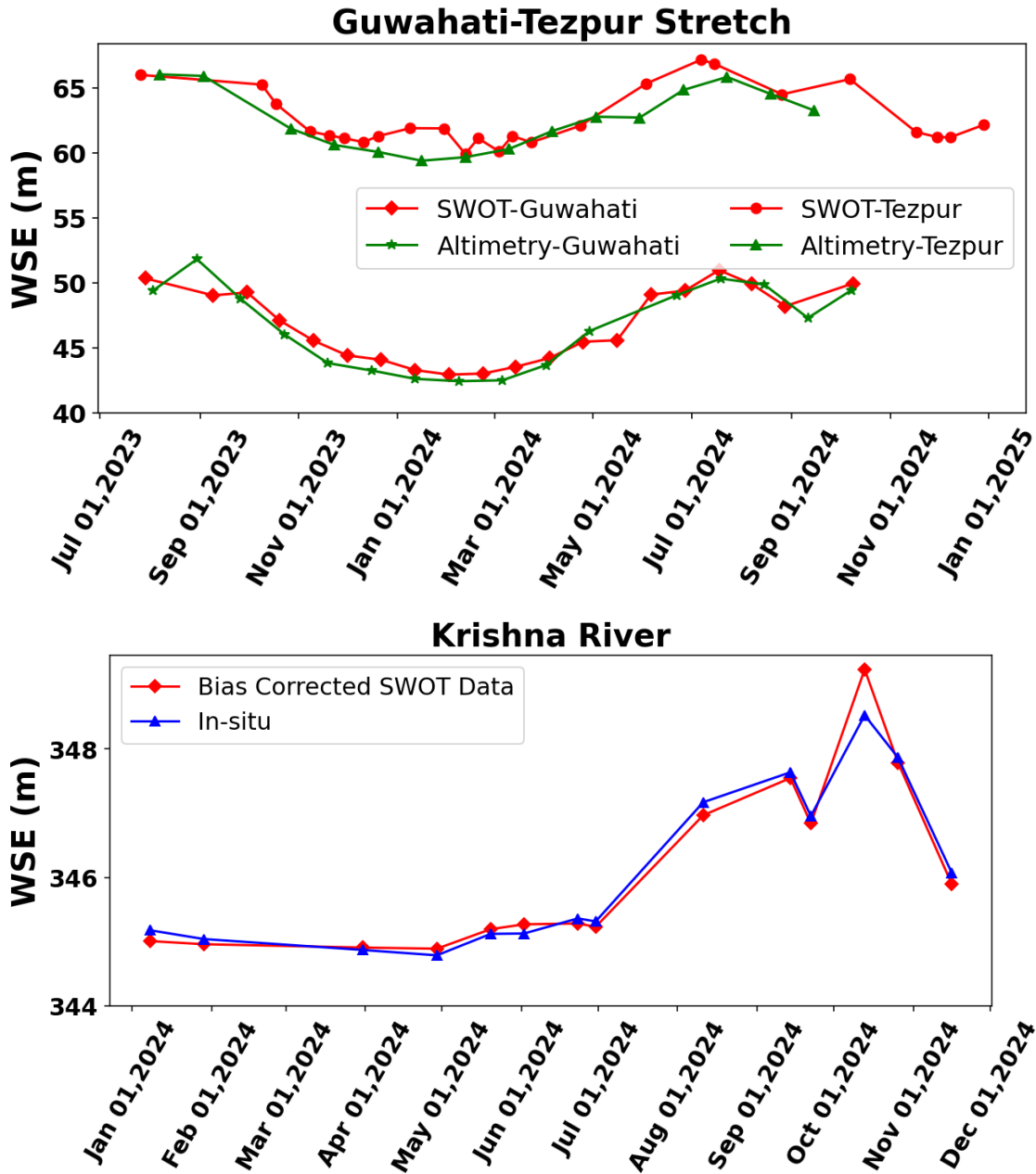


Hirakud Reservoir



Kedartal





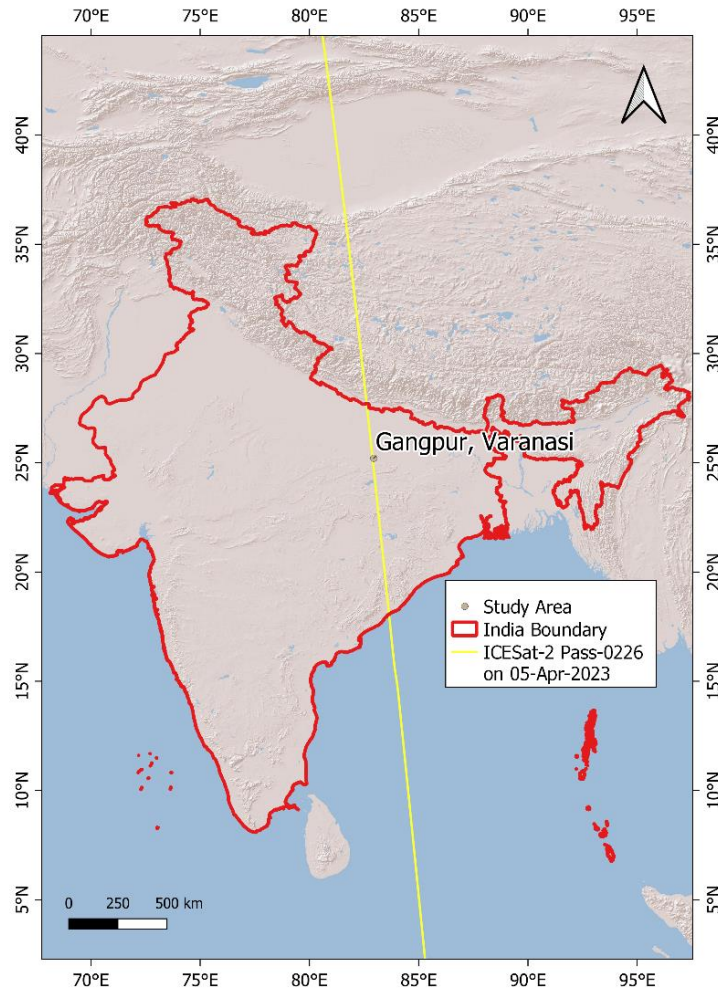
3.2.2. Bathymetry estimation:

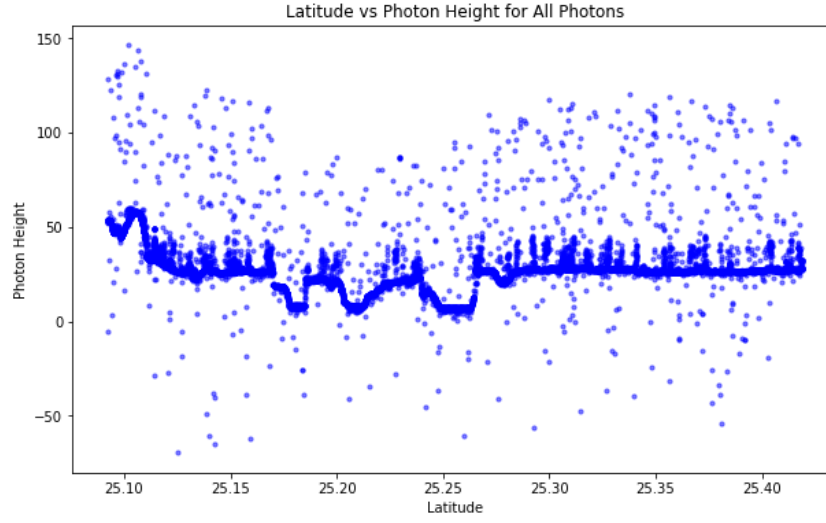
The overarching goal of this study is to develop and validate a methodology for estimating river discharge at virtual stations using remote sensing and hydrodynamic modelling. The specific objectives include:

- Estimating river cross-sections from satellite-derived water surface elevations and widths.

- Predicting unknown river bathymetry using the Non-Linear Slope Break (NLSB) model.
- Developing an automated algorithm for cross-section (bathymetry) extraction and discharge estimation.

Validating the methodology for the Ganga river and extending it to other river basins for broader applicability





The methodology integrates satellite data processing, statistical modeling, and hydrodynamic analysis to estimate river discharge. The workflow consists of several key steps:

i. Non-Linear Slope Break (NLSB) Model

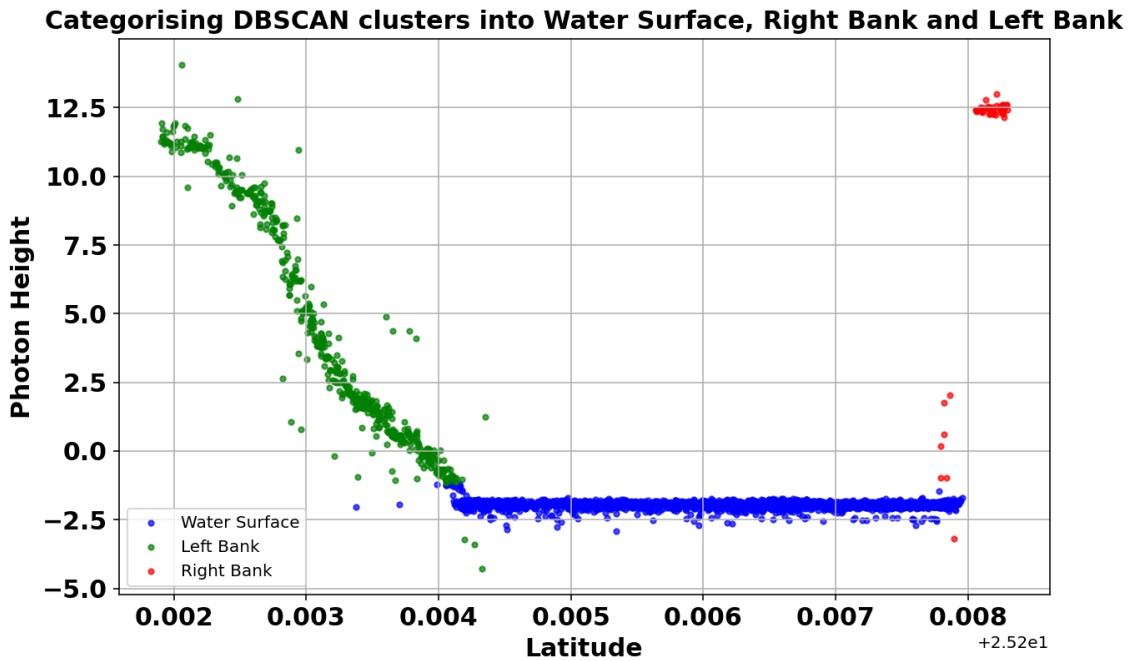
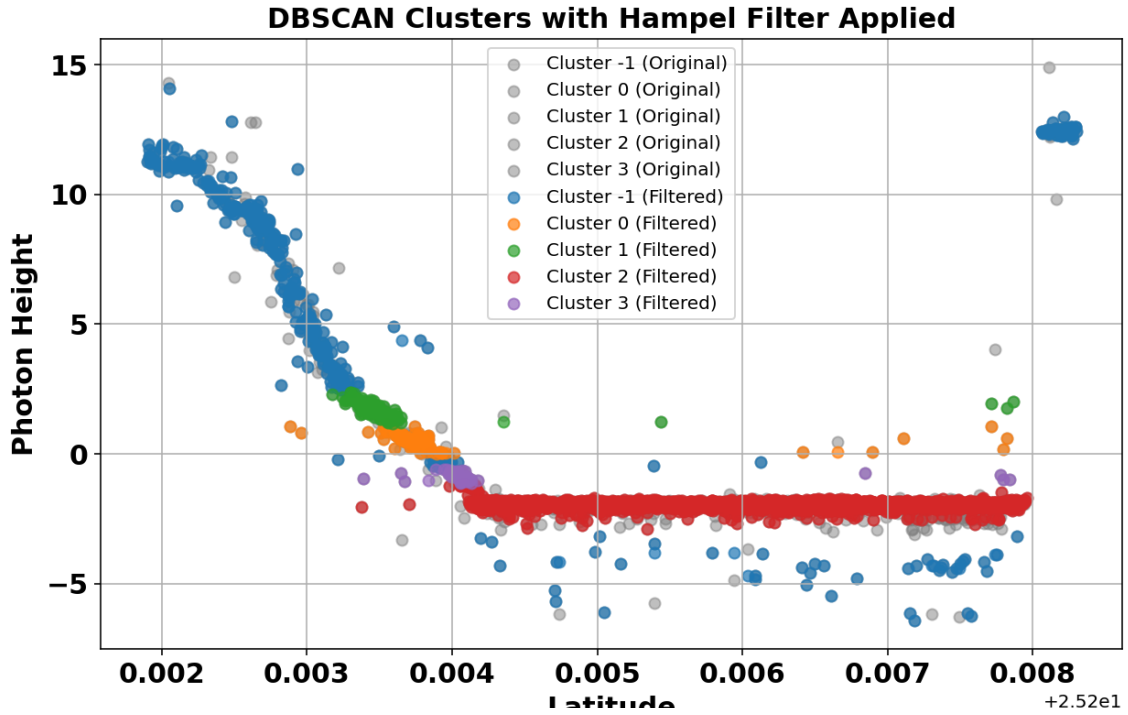
The NLSB model, as described by Schaperow et al. (2019), is a curve-fitting approach used to estimate unknown river bathymetry from observed water surface height and width data. The model assumes that the relationship between water surface elevation and river width can be described by a non-linear function, which, when fitted to the data, reveals the underlying channel geometry. This approach is particularly useful when direct bathymetric measurements are unavailable, as is often the case at virtual stations.

$$h = \begin{cases} z_0 + u_1 w^{s_1}, & \text{for } w \leq w_b \\ z_0 + u_1 w_b^{s_1} + u_2 (w^{s_2} - w_b^{s_2}), & \text{for } w \geq w_b \end{cases}$$

a. Automated Algorithm Development

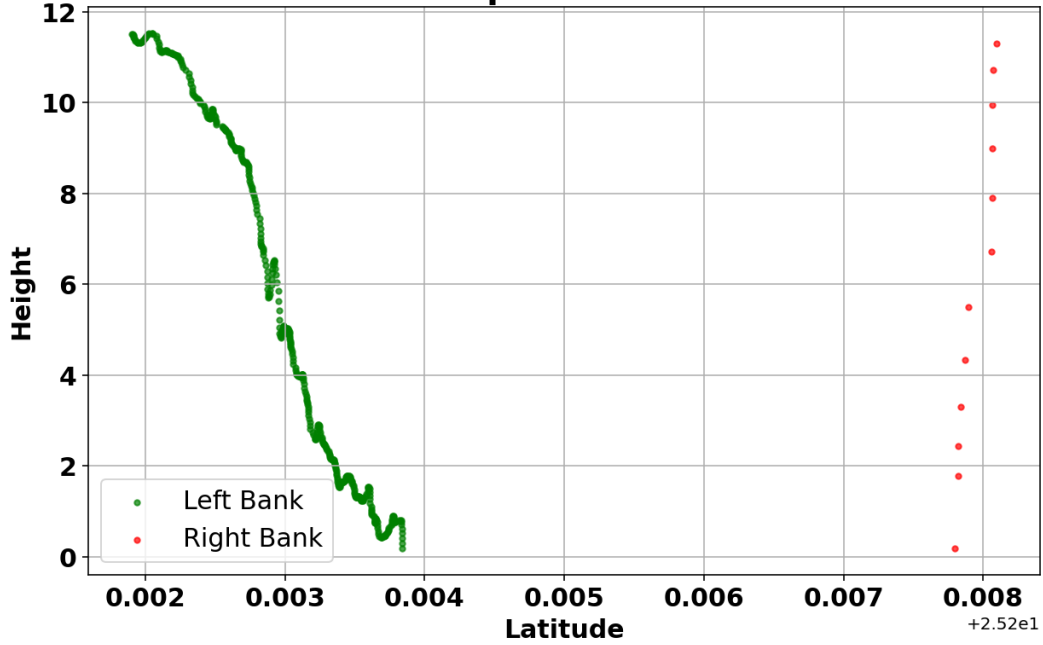
To operationalize the NLSB model and streamline the estimation process, an automated algorithm is being developed. The main components of the algorithm include:

- **Section** **Selection:**
A brute force method is used to select representative river sections for analysis. This ensures that the chosen sections capture the variability in channel morphology and flow conditions.
- **DBSCAN** **Clustering:**
The Density-Based Spatial Clustering of Applications with Noise (DBSCAN) algorithm is applied to the elevation data to identify the water surface. DBSCAN is effective at distinguishing dense clusters (e.g., water surface points) from noise (e.g., outliers or non-water points), facilitating accurate surface identification.
- **Hampel** **Filtering:**
After clustering, Hampel filtering is employed to remove outliers and noise from the water surface data. This robust statistical filter enhances data quality by preserving true signal features while eliminating anomalous points.
- **Categorization:**
The processed data are categorized into four key classes: water surface, right bank, left bank, and river bed. This classification is essential for accurately delineating the river cross-section and fitting the NLSB model.
- **Model** **Fitting:**
The function *nlsb_fit* is used to adjust the parameters of the NLSB model. The algorithm iteratively searches for the parameter set that minimizes the error between observed and predicted values, ensuring the best possible fit to the data.
- **Monte Carlo** **Simulations:**
To enhance the robustness of the model, Monte Carlo simulations are incorporated. These simulations involve generating multiple realizations of the input data and model parameters, allowing for uncertainty quantification and improved prediction reliability.



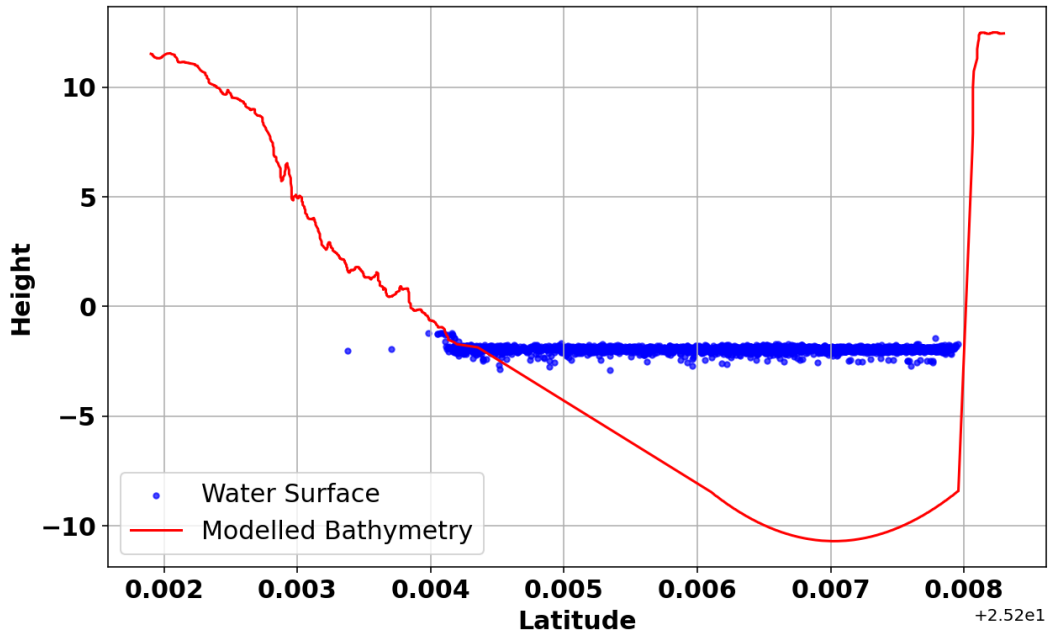
Gangpur Varanasi (25.21, 82.93)

WSEw pair extraction



Gangpur Varanasi (25.21, 82.93)

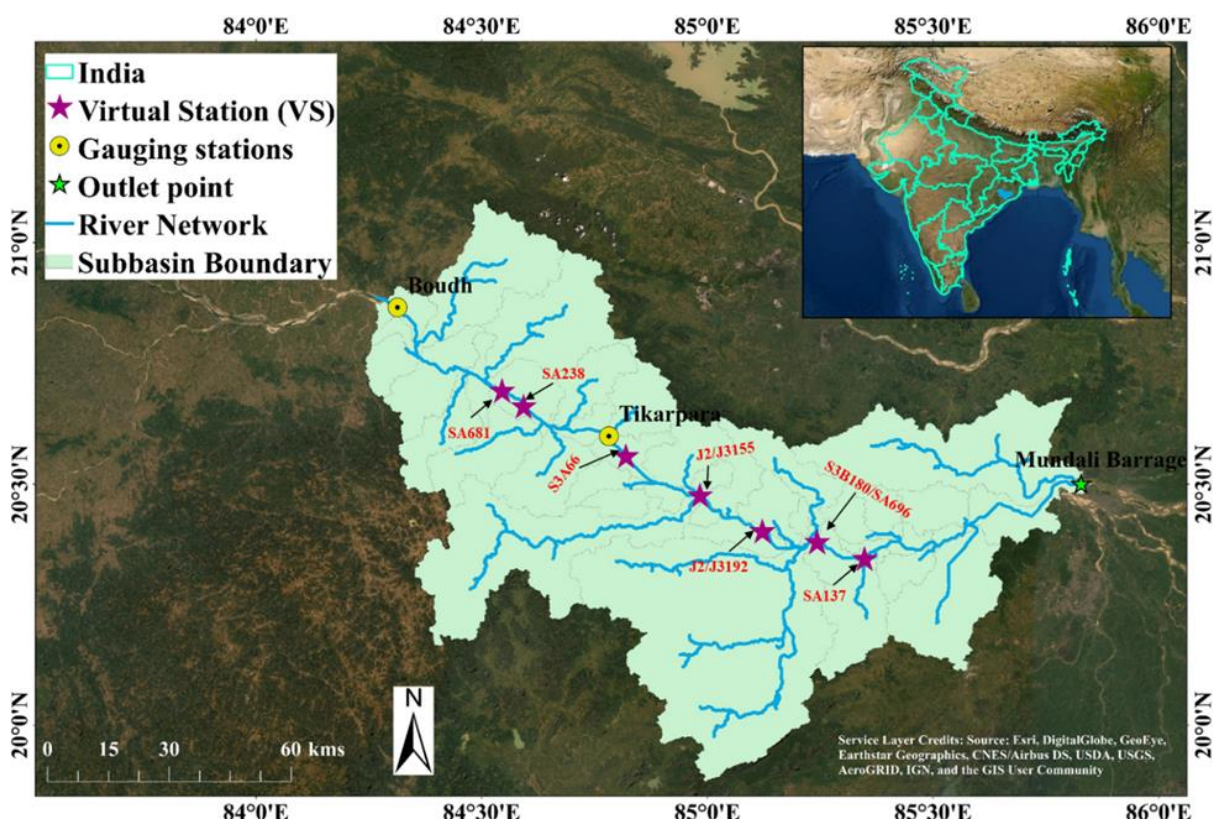
Predicated River Cross-Section



3.2.3. Discharge Estimation:

The limited availability of in-situ data has drawn attention towards using remote sensing techniques to monitor river discharge. Radar altimetry data has been used to generate stage-discharge rating curves through power-law relations and empirical methods. However, evaluating hydrodynamic models for rating curve generation using multi-mission altimetry data in data-scarce regions is lacking. We used altimetry data (Jason 2, Jason 3, SARAL/AltiKa, Sentinel 3A, and Sentinel 3B) over the Mahanadi River to evaluate rating curves at virtual stations.

The HEC-RAS hydrodynamic model was set up, identifying seven virtual stations along the Mahanadi River from Boudh to Mundali Barrage. The availability of rating curves at virtual stations allows the expansion of the gauging network along the sparsely gauged river basin. This approach provides the low-cost uniform monitoring of river flow in remote areas.



3.2.4. Field Visit:

A. Vasudhara Glacial Lake

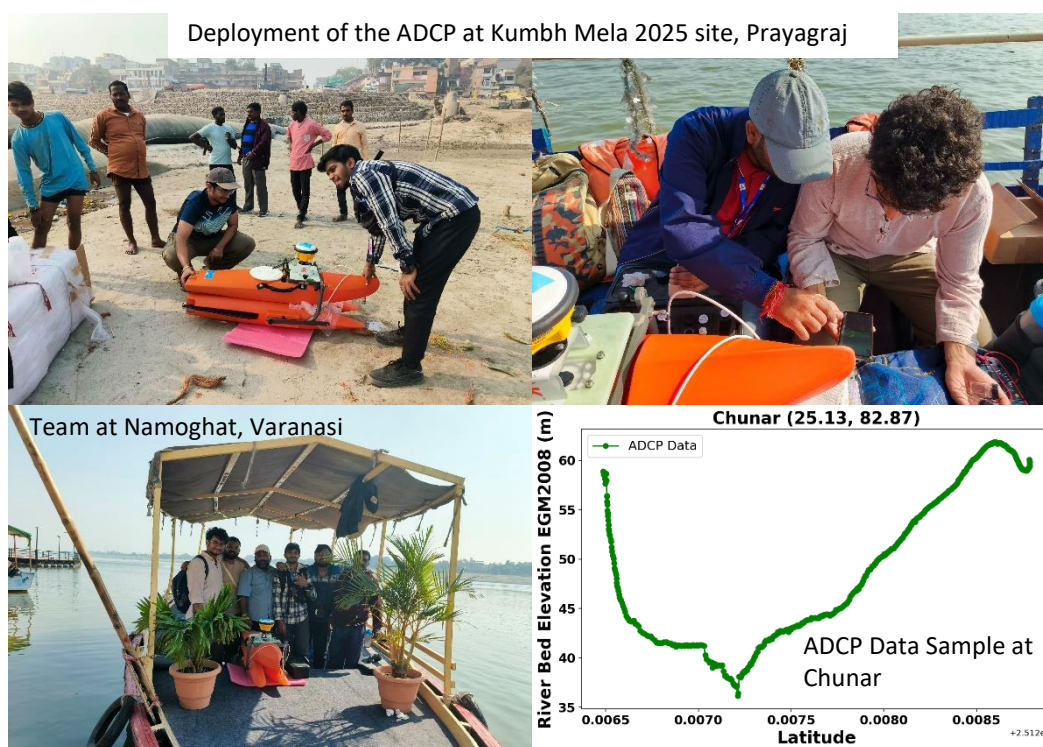
Bathymetry survey of the Vasudhara Glacial Lake for lake volume estimation



B. Ganga River Stretch from Kanpur to Varanasi

A field visit was carried out from 30th November to 07th December 2024 in Ganga River Stretch from Kanpur to Varanasi. The main objective of the field of the was to collect the validation data from the ADCP for the bathymetry estimation model at total of 12 sites.

Fig. 2. Field Images

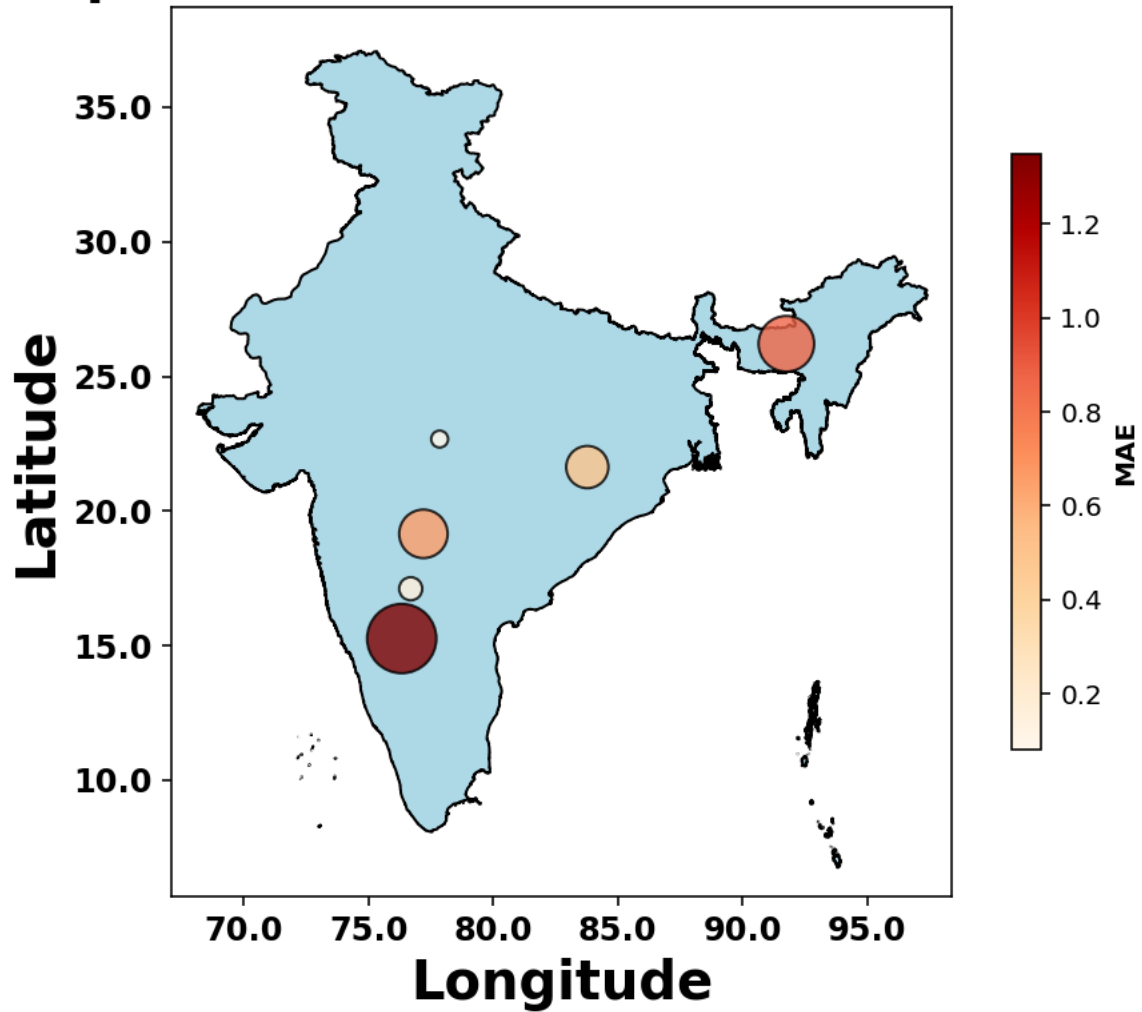


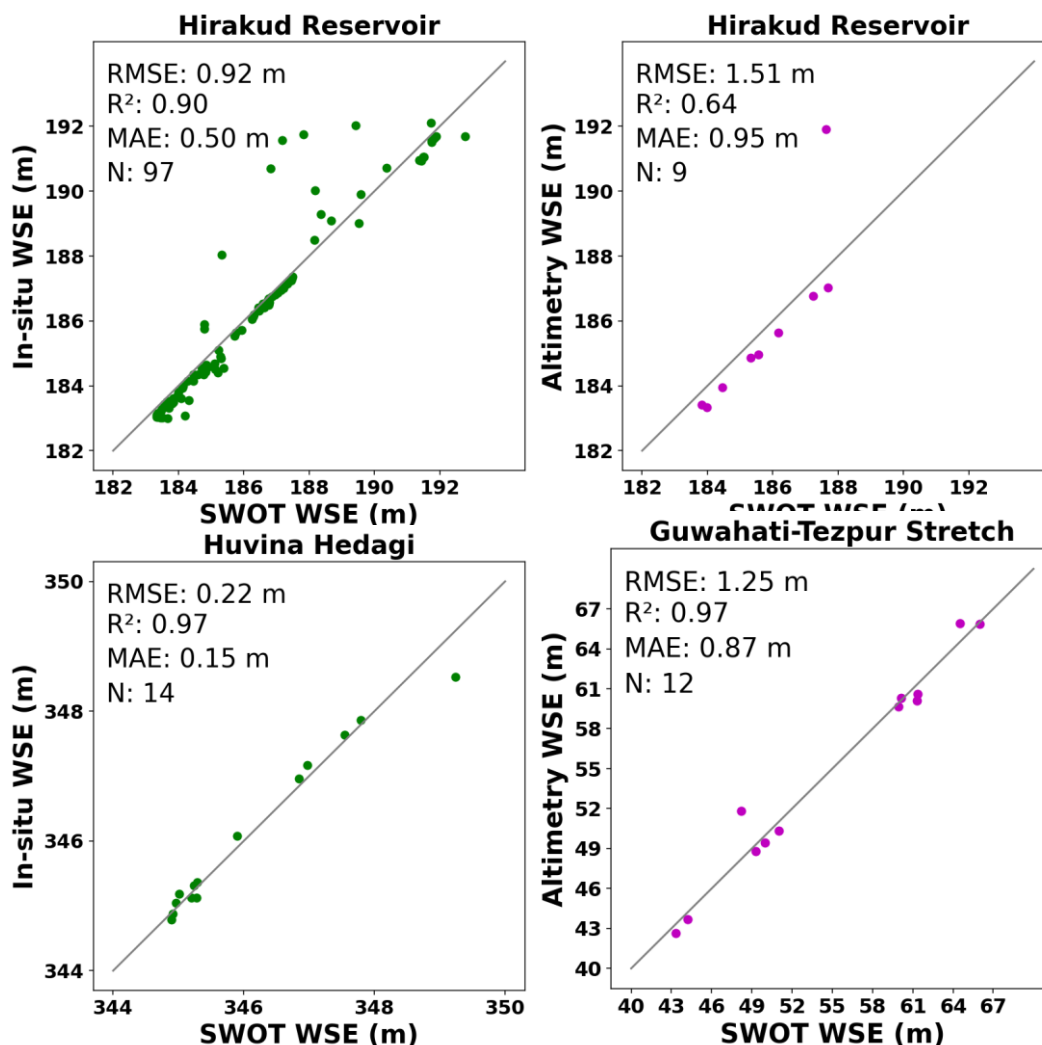
4. RESULTS:

4.1. Water Level Extraction:

Validation of the SOWT derived WSE from the insitu and historical altimetry data. The MAE for different site have been illustrated in the figure.

Spatial Distribution of MAE values



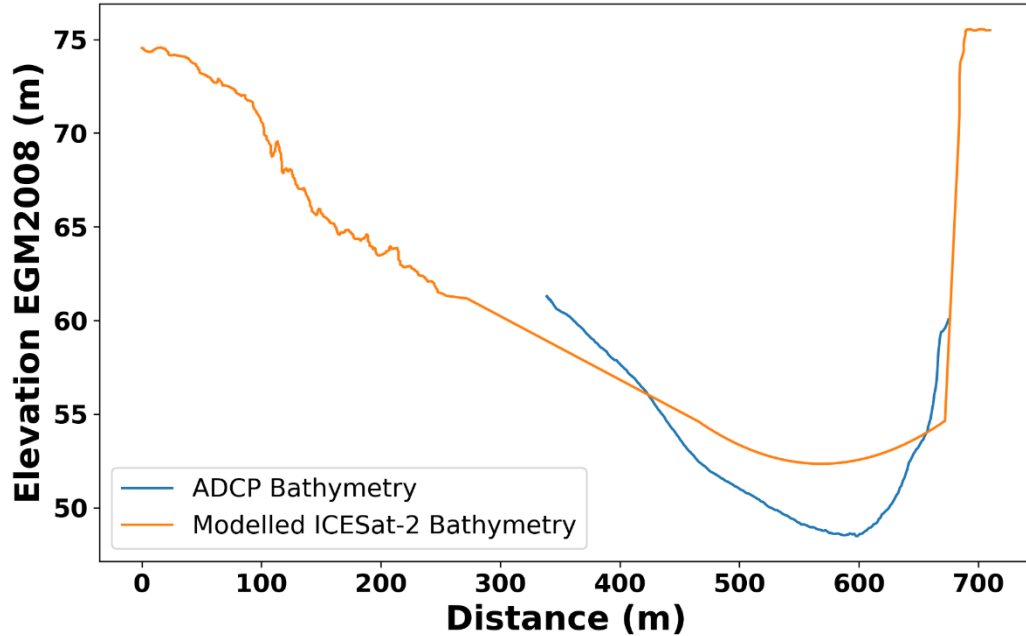


4.2. Bathymetry Estimation:

Validation involves comparing model predictions with available in-situ measurements assessing accuracy and reliability.

Error metrics such as RMSE 2.40 m indicate reasonable accuracy across 288 data points.

Gangpur Varanasi (25.21, 82.93)
Date-20230405 Pass-0226 Beam-3



Site	Latitude	Longitude	Date	ICESat-2 Pass	Beam	RMSE	MAE
Gangpur Varanasi	[25.0921	[82.93670	20230405	226	3	2.403446627	2.158377542

Relative Maximum Depth Error	Relative Maximum Depth Error %	Relative Low Flow Area Error	Relative Low Flow Area Error %	Low Flow Area Percentage for the modelled bathymetry	Sample Size
4.054587857	14.98243181	568.7684496	57.02905141	4.301014138	288

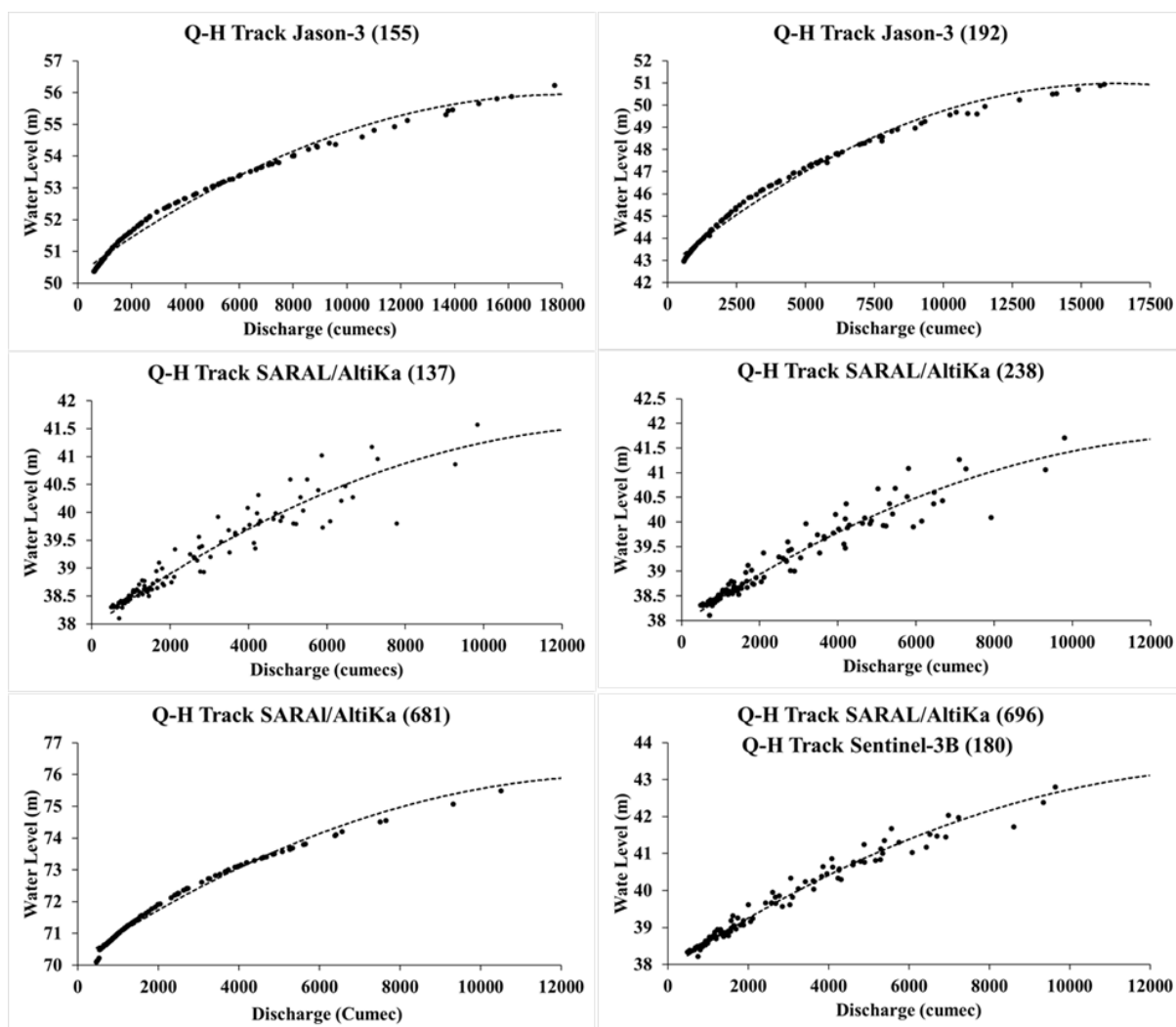
Significant progress has been made in developing and testing the automated algorithm:

- The integration of the NLSB model with Monte Carlo simulations has been successfully implemented, providing a framework for robust bathymetry and discharge estimation.
- The application of DBSCAN clustering and Hampel filtering has demonstrated effective identification and cleaning of water surface data, a critical step for accurate cross-section extraction.
- The categorization process has enabled clear distinction between water surface, banks, and river bed, facilitating precise model fitting.
- The *nlsb_fit* function has yielded promising results, with adjusted parameters minimizing error and improving the fidelity of bathymetric predictions.

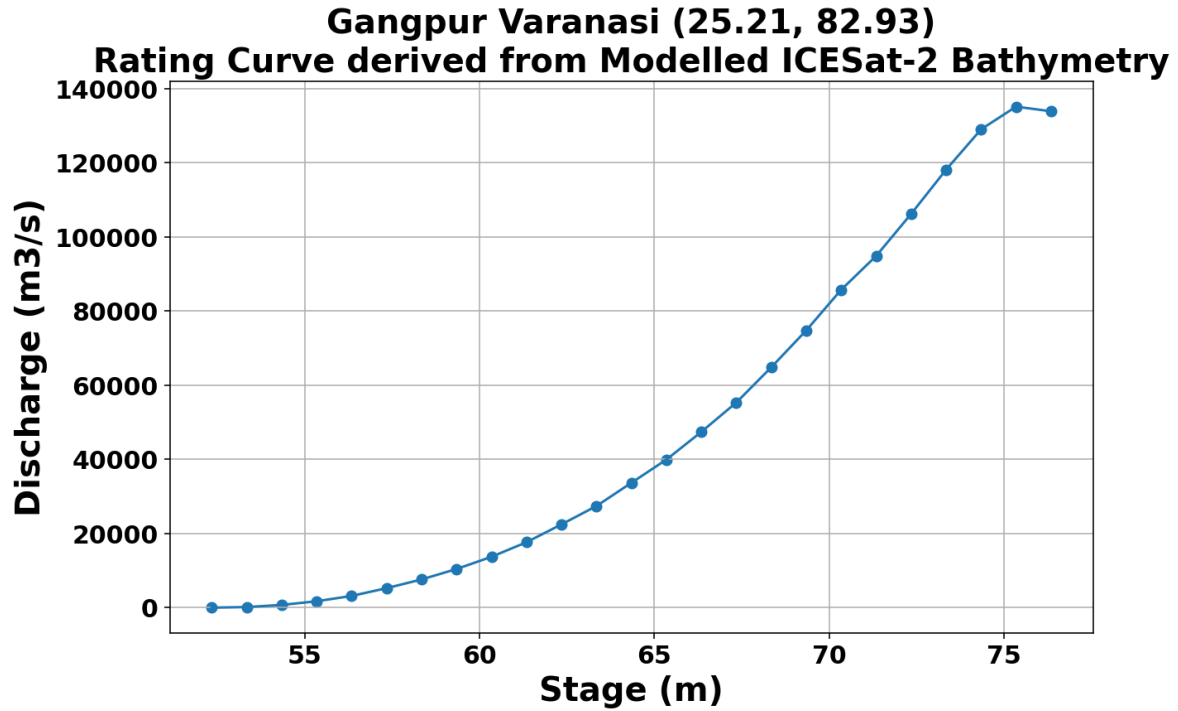
- Preliminary validation indicates that the methodology can reliably estimate river cross-sections and discharge at virtual stations, even in the absence of direct measurements.

4.3. Discharge Estimation:

- A. Statistical evaluation of 'water level' (RMSE 0.27-0.88 m) and 'discharge' (KGE 0.52 to 0.88) components of the generated rating curves showed significant agreement with radar altimetry data.



- B. The hydraulic and statistical summaries for Gangpur Varanasi (ICESat-2 Pass 226, Beam 3) show strong model performance, with a modelled discharge of 706.87 m³/s closely matching the observed (ADCP) 665 m³/s. Error metrics such as RMSE 2.40 m indicate reasonable accuracy across 288 data points.



Site	Latitude	Longitude	Date	ICESat-2 Pass	Beam	Water Level in m (EGM2008)
Gangpur Varanasi	[25.0921	[82.93670	20230405	226	3	55.25173755

Wetted Area in m2	Wetted Perimeter in m	Hydraulic Radius in m	Mannings n	S	Q_modelled in m3/s	Q_ADCPin m3/s
428.562622	732.0974693	0.585390115	0.03	0.005	706.8733815	665

5. SUMMARY:

The integration of satellite remote sensing and hydrodynamic modeling represents a transformative advance in river discharge estimation. By leveraging high-resolution ICESAT-2 data and sophisticated statistical models, we can overcome the limitations of traditional in-situ measurements, providing continuous, spatially extensive discharge estimates.

The use of the NLSB model is particularly advantageous for virtual stations, where bathymetric data are typically unavailable. The automated algorithm, combining clustering, filtering, and model fitting, ensures scalability and reproducibility, making the approach suitable for large-scale hydrological monitoring.

However, several challenges remain. The accuracy of discharge estimates depends on the quality of satellite data, the appropriateness of the model assumptions, and the representativeness of the selected river sections. Variability in channel morphology, vegetation cover, and water quality can introduce uncertainties that must be carefully managed. Ongoing validation and refinement, supported by ground truth data were available, are essential for operational deployment.

Future work will focus on extending the methodology to additional river basins, enhancing algorithm automation, and conducting comprehensive validation studies. By advancing the state of river discharge estimation, this approach holds promise for improving water resource management, flood forecasting, and environmental monitoring on a global scale.

6. PUBLICATIONS:

- “SWOT Mission with Wide-Swath Altimetry: Observations and Insights into India’s Inland Waterbodies,” Link: <https://link.springer.com/article/10.1007/s12524-025-02234-8>
- <https://science.iirs.gov.in/observing-the-changes-in-inland-water-bodies-through-satellite-based-wide-swath-altimetry-mission/>
- “Nadir and Wide Swath Altimetry for Inland Water Bodies”, Technical Report (Submitted)

7. REFERENCES

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- Dhote, P.R., Agarwal, A., Singhal, G., Calmant, S., Thakur, P.K., Oubanas, H., Paris, A., Singh R.P. 2024. River Water Level and Water Surface Slope Measurement From Spaceborne Radar and LiDAR Altimetry: Evaluation and Implications for Hydrological Studies in the Ganga River. IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, vol. 17, pp. 7825-7840, 2024, <https://doi.org/10.1109/jstars.2024.3379874>.
- Durand, Michael, Gleason, C. J., Pavelsky, T. M., Prata de Moraes Frasson, R., Turmon, M., David, C. H., et al. (2023). A Framework for Estimating Global River Discharge From the Surface Water and Ocean Topography Satellite Mission. Water Resources Research, 59(4). <https://doi.org/10.1029/2021WR031614>
- Getirana, A., Kumar, S., Bates, P. et al. The SWOT mission will reshape our understanding of the global terrestrial water cycle. Nat Water 2, 1139–1142 (2024). <https://doi.org/10.1038/s44221-024-00352-0>

8. CONTRIBUTION TO ACADEMIC ACTIVITIES OF DEPT/ GROUP:

a) Institutional Activities:

- a. Participated in Elocution Competition on October 7, 2024
- b. Participated in One-day workshop on “Data Privacy & Authenticity in Cyber Space” on November 8, 2024.
- c. Committee Member at the Hindi Technical Sangosthi January 10, 2025
- d. 2 Lectures Delivered in IIRS Lecture Series ‘Vyakhyan’ namely Principal Component Analysis – November 22, 2024 and Estimating PI with Buffon’s needle – March 19, 2025.
- e. “Water Rocket Demonstration” during YUVIKA 2025.

b) Invigilation Duties during M.Tech/PGD examinations

c) Department Activities:

- a. Practical conducted during the departmental courses of CWC Cadre IES officials, ITEC, NNRMS, Rajasthan Government Officials
- b. Field Visit during the departmental courses for instrument demonstrations of ITEC at Asan Barrage and Dak Pathar

d) Tool Creation: Automated water level extraction for the major Indian river basin from DAHITI website.